SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Criteria	**Problem Interpretation (2)**	**Analytical Reasoning (2.5)**	**Method Selection (2)**	**Solution Quality (2)**	**Presentation of Analysis (1.5)**	**TOTAL (10)**
Weightage	**20%**	**25%**	**20%**	**20%**	**15%**	**100%**
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty Total Marks Awarded:

Analytical Problem Solving

1. A cloud datacenter host has 64 CPU cores and supports CPU overcommitment at a ratio of 2.5:1. Each virtual machine (VM) requires 4 vCPUs.Calculate the maximum number of VMs that can be deployed on this host. If actual CPU utilization reaches 85%, analyze whether further VM allocation is advisable.
2. A virtualized storage system provides 20 TB of physical storage with thin provisioning at a ratio of 3:1. Each VM is allocated 500 GB of virtual disk space. Calculate the maximum number of VMs that can be provisioned.
3. Evaluate the security implications of multi-tenancy in cloud datacenters. Analyze how virtualization can both enable and weaken isolation, and recommend mechanisms to strengthen tenant separation.
4. Analyze the role of Service-Oriented Architecture (SOA) in enabling interoperability across heterogeneous cloud platforms. Identify key challenges such as service latency, security, and version control, and propose solutions.
5. A cloud system implementing presence services faces scalability issues during peak usage. Analyze the architectural bottlenecks and recommend design improvements to ensure real-time updates and efficient resource utilization.

1.Cloud Datacenter CPU Overcommitment

Given:

- Physical CPU cores = **64**
- CPU overcommitment ratio = **2.5 : 1**
- Each VM requires **4 vCPUs**

Step 1: Calculate Total Available vCPUs

$$64 \times 2.5 = 160 \text{ vCPUs}$$

Step 2: Calculate Maximum Number of VMs

$$\frac{160}{4} = 40 \text{ VMs}$$

Answer:

The maximum number of VMs that can be deployed is **40 VMs**.

Analysis at 85% CPU Utilization

- At **85% CPU utilization**, the server is already heavily loaded.
- Deploying additional VMs may:
  - Increase CPU contention.
  - Reduce application performance.
  - Cause slower response times.
  - Affect Quality of Service (QoS).

Conclusion

Although the host can theoretically support **40 VMs**, allocating more VMs when CPU utilization has reached **85% is not advisable**. Administrators should either migrate workloads, add another host, or increase physical resources.

2. Virtualized Storage with Thin Provisioning

Given:

- Physical storage = **20 TB**
- Thin provisioning ratio = **3 : 1**
- Virtual disk per VM = **500 GB**

Step 1: Calculate Virtual Storage

$$20 \times 3 = 60 \text{ TB}$$

Step 2: Convert TB to GB

$$60 \times 1000 = 60000 \text{ GB}$$

Step 3: Calculate Maximum Number of VMs

$$\frac{60000}{500} = 120$$

Answer
The maximum number of VMs that can be provisioned is **120 VMs**.

Advantages of Thin Provisioning

- Better storage utilization.
- Lower storage cost.
- Easy VM deployment.
- Flexible resource allocation.

Risks

- Storage over-allocation.
- Performance degradation.
- Risk of running out of physical storage.

Conclusion
Thin provisioning improves storage efficiency, but administrators must continuously monitor actual storage usage to avoid storage exhaustion.

3. Security Implications of Multi-Tenancy in Cloud Datacenters

Multi-Tenancy
Multi-tenancy allows multiple customers (tenants) to share the same physical hardware while keeping their applications and data isolated.

Security Benefits

- Efficient resource utilization.
- Lower infrastructure cost.
- Easy scalability.
- Centralized management.

Security Risks

1. Data leakage between tenants.
2. Hypervisor attacks.
3. Side-channel attacks.
4. VM escape vulnerabilities.
5. Insider threats.

How Virtualization Improves Isolation

- Separate virtual machines.
- Memory isolation.
- Virtual networking.
- Independent operating systems.

Recommended Mechanisms

- Strong hypervisor security.
- Encryption of stored and transmitted data.
- Multi-factor authentication (MFA).
- Role-Based Access Control (RBAC).
- Network segmentation.
- Continuous security monitoring.
- Regular security updates and patching.

Conclusion

Virtualization enhances resource sharing and isolation, but strong security controls are essential to protect tenants from unauthorized access and cyber threats.

4. Role of Service-Oriented Architecture (SOA) in Cloud Platforms

Service-Oriented Architecture (SOA)

SOA is an architectural approach where applications are divided into reusable and independent services that communicate through standard protocols.

Importance of SOA

- Promotes interoperability.
- Enables service reuse.
- Simplifies integration.
- Supports scalability.
- Facilitates cloud migration.

Challenges

1. Service Latency

- Network delays
- Slow response during peak loads

2. Security

- Unauthorized access
- Data interception
- Identity management issues

3. Version Control

- Different service versions
- Compatibility problems

Solutions

- API Gateway implementation.
- Load balancing.
- Service caching.

- OAuth and JWT authentication.
- Encryption using TLS.
- Version management.
- Microservices architecture.
- Continuous monitoring.

Conclusion

SOA enables seamless communication between heterogeneous cloud platforms. Proper management of latency, security, and version control ensures reliable and efficient cloud services.

5. Presence Service Scalability in Cloud Systems

Presence Service

A presence service continuously updates users' online or offline status in real time.

Architectural Bottlenecks

1. High network traffic.
2. Centralized server overload.
3. Database bottlenecks.
4. Frequent status updates.
5. Increased memory consumption.

Recommended Design Improvements

Horizontal Scaling

- Add multiple servers to distribute workload.

Load Balancing

- Evenly distribute user requests.

Distributed Databases

- Replicate data across multiple servers.

Message Queues

- Process updates asynchronously.

Caching

- Store frequently accessed data using Redis or Memcached.

Event-Driven Architecture

- Trigger updates only when status changes.

Auto Scaling

- Automatically increase or decrease cloud resources based on demand.

Benefits

- Improved scalability.
- Reduced latency.
- Better resource utilization.
- High availability.
- Faster real-time updates.

Conclusion

Cloud presence services can efficiently handle peak loads by adopting distributed architecture, load balancing, caching, and auto-scaling. These improvements ensure real-time communication while maximizing resource utilization.